

Mixed Mode Surveys Including Self-Administration in Interviews

SurvMeth/Surv 621
November 18, 2024

Mixed-Mode Survey Design

- Researchers commonly combine survey modes in order to
 - Increase RRs, increase coverage, improve data quality
 - Optimize **trade-off** between different costs and errors
- Concern is creation of *mode effects*
 - different patterns of data between modes that are hard to attribute to anything other than mode
- Beyond collecting data, different modes can be used to contact/invite and communicate with *Rs*
 - e.g., survey invitation in one mode, administration of questionnaire in another

Principle of Hypothesis Testing

- Example Hypothesis:
 - For all potential survey respondents, if multiple survey modes are provided, then they will be more likely to participate in the survey.
 - The empirical test operationalizes each component into a concrete realization:
 - All potential survey respondents -> Specific sample persons
 - Survey mode -> Some set of survey modes
 - Propensity to participate -> Actual participation decisions in specific surveys
 - Hypothesized causal link -> Observed covariation in sample

Hypothesis Testing and Validity to Causal Inference

- What are the potential inferential problems?
 1. Is there a large and reliable relationship between the cause and effect?
 2. Is the relationship causal?

Covariation: When the values on one variable change, so do values on the other variables change, they are “co-related”

Temporal Order: Changes on the independent variable precede changes on the dependent variable in time. X is the cause of Y and not the reverse.

Elimination of rival hypotheses: Other potential causes are accounted for and ruled out. (i.e. plausible alternative hypotheses are eliminated)
 3. Does it involve the theoretical variables of interest?
 4. How generalizable is the relationship over samples, settings, methods, etc?

Hypothesis Testing and Validity to Causal Inference

- In the tradition of experimental design (Cook & Campbell; Campbell & Stanley), these four issues are expressed with the following terms:
 - 1 Statistical conclusion validity
 - 2 Internal validity
 - 3 Construct validity
 - 4 External validity

Mixed-Mode Systems

- Contact Phase Mode Change
- Response Phase Mode Change
- Follow-up Phase Mode Change

Contact Phase Mode Change

Mixed-Mode Survey System	Rationale for Implementation Quality	Effect on Survey
Contact Phase Mode Change		
Advance notification in different mode than data collection	• Correct sampling frame • Raise response • Enhance credibility/trust	• Reduce coverage and nonresponse error • No threats to measurement if data collection single-mode
Recruitment/ Screening/ Selection in different mode than data collection	• Reduce cost • Enhance efficiency • Update/ expand contact information for main code	• Timeliness • If pure screening, no threats to measurement • If screening plus first part data collection in other mode potential mode effects on measurement

Contact Phase Mode Change

<i>Mixed-Mode Survey System</i>	<i>Rationale for Implementation Quality</i>	<i>Effect on Survey</i>
Recruitment/ Screening/ Selection in different mode than data collection	• Reduce cost • Enhance efficiency • Update/ expand contact information for main code	• Timeliness • If pure screening, no threats to measurement • If screening plus first part data collection in other mode potential mode effects on measurement

Examples

- Link, M. W., & Lai, J. W. (2011). Cell-Phone-Only Households and Problems of Differential Nonresponse Using an Address-Based Sampling Design. *Public Opinion Quarterly*, 75(4), 613–635.

Table 2. Household Telephone Access by Sample Type

Regular Sample 87,034 (100%)							
Matched Sample 60,346 (69.3%)				Unmatched Sample 26,688 (30.7%)			
Cell Only	Landline & Cell	Landline Only	No Tele	Cell Only	Landline & Cell	Landline Only	No Tele
2,167 (2.5%)	53,408 (61.4%)	4,565 (5.2%)	206 (0.2%)	10,159 (11.7%)	15,063 (17.3%)	1,301 (1.5%)	165 (0.2%)

DATA SOURCE.—Spring 2009 Radio Audience Measurement.

Response Phase Mode Change

Mixed-Mode Survey System	Rationale for Implementation Quality	Effect on Survey
Response Phase Mode Change		
Different (sample) persons by different modes when surveying one sample at one time period with one questionnaire	• Reduce cost • Improve coverage • Improve response	• Reduce coverage and nonresponse error • Mode effects on measurement confounded with subgroups

Examples

- Cobben, F., Schouten, B., & Bethlehem, J. (2006). A Model for Statistical Inference based on Mixed Mode Interviewing. In European Conference on Quality in Survey Statistics. Retrieved from http://www.statistics.gov.uk/events/q2006/downloads/T10_Cobben.doc
- Beebe, T. J., McAlpine, D. D., Ziegenfuss, J. Y., Jenkins, S., Haas, L., & Davern, M. E. (2012). Deployment of a Mixed-Mode Data Collection Strategy Does Not Reduce Nonresponse Bias in a General Population Health Survey. *Health Services Research*, 47(4), 1739–1754.
- Baines, A., Partin, M., & Davern, M. (2007). Mixed-mode administration reduced bias and enhanced poststratification adjustments in a health behavior survey. *Journal of Clinical Epidemiology*, 1246–1255. Retrieved from <http://linkinghub.elsevier.com/retrieve/pii/S0895435607000704>
- Ainsaar, M., Lilleoja, L., Lumiste, K., & Roots, A. (2013). ESS Mixed Mode Experiment Results in Estonia (CAWI and CAPI mode sequential design). Retrieved from http://www.ssi.ut.ee/sites/default/files/ssi/ess_dace_mixed_mode_ee_report.pdf
- Millar, M. M., & Dillman, D. A. (2011). Improving Response to Web and Mixed-Mode Surveys. *Public Opinion Quarterly*, 75(2), 249–269.

Response Phase Mode Change

Mixed-Mode Survey System	Rationale for Implementation Quality	Effect on Survey
Different parts of a questionnaire by different modes when surveying one sample at one time point	• Improve privacy of measurement • Reduce social desirability	• Improved data quality, especially with very sensitive questions

Examples

- Jones, R. K., & Kost, K. (2007). Underreporting of Induced and Spontaneous Abortion in the United States: An Analysis of the 2002 National Survey of Family Growth, 187–197.
- Tourangeau, R., & Smith, T. W. (1996). Asking Sensitive Questions the Impact of Data Collection Mode, Question Format, and Question Context. *Public Opinion Quarterly*, 60, 275–304.

Response Phase Mode Change

Mixed-Mode Survey System	Rationale for Implementation Quality	Effect on Survey
Same person with different modes at multiple time points (panel)	• Reduce costs	• Measurement differences causing confounding of time and mode effects

Examples

- Peracchi, F., & Welch, F. (1995). How representative are matched cross-sections? Evidence from the Current Population Survey. *Journal of Econometrics*, 68, 153–179.

Response Phase Mode Change

Mixed-Mode Survey System	Rationale for Implementation Quality	Effect on Survey
Different (whole) samples by different modes, often at different times with different questionnaires	• Comparative research • Different research traditions • Different coverage • Different cost structure	• Coverage error • Nonresponse error • Measurement error • Incomparability

Examples

- Martin, P. (2011). What makes a good mix? Chances and challenges of mixed mode data collection in the ESS (No. 2). Retrieved from https://www.city.ac.uk/__data/assets/pdf_file/0015/125133/CCSS-Working-Paper-No-02.pdf
- HaileMichael, G. W. (2013). Quality assurance and quality control in multinational and multicultural surveys. Stockholm University.

Follow-up Phase Mode Change

Mixed-Mode Survey System	Rationale for Implementation Quality	Effect on Survey
Follow-up Phase Mode Change		
Reminders in different modes from mode in which all respondents are asked to complete questionnaire	• Raise response	• Reduce nonresponse error • If pure reminder no threats to measurement • If reminder plus part data collection in other mode risk of potential mode effects on measurement

Outline

- Mixed Mode Surveys
 - Pre-notification and Invitation
 - Data collection: Sequential vs. Concurrent designs
- Mode Choice
 - vs. mode preference
- Switching modes (to self-administration) within an interview
 - In person: Effects of audio, computerization
 - Phone: IVR

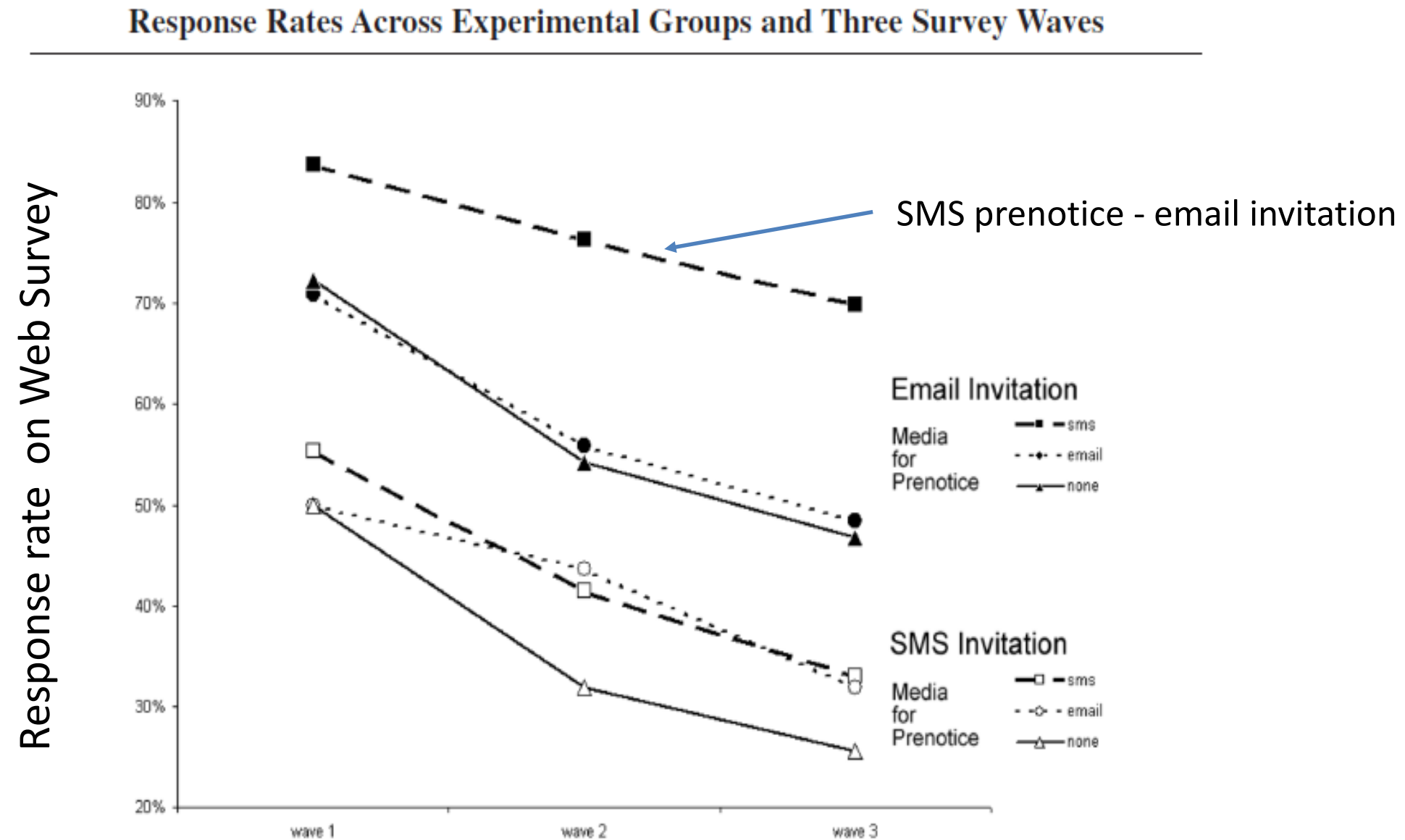
Mixed-Mode Design: *Pre-notification*

- Different modes in *pre-notification* than *data collection*
 - Pre-notification alerts sample member to an upcoming survey invitation – not actually an invitation
- Examples:
 - Postal mail pre-notification for Web survey
- Goals:
 - Increase response rate by increasing study's salience

Mixed-Mode Design: *Pre-notification* and *Invitation*

- Bosnjak et al. (2008) tested impact on Web survey RR using three pre-notification (SMS, email, none) and two invitation (SMS, email) modes
- Nearly all participants owned a mobile phone
 - SMS most effective pre-notification mode
 - Email more effective invitation mode
 - SMS pre-notification + email invitation most successful mode combination for Web survey

Mode of Pre-notification and Invitation



Bosnjak et al. (2008)

Mixed-Mode Design: *Data Collection*

- Concurrent mixed-mode design: One sample, cross-sectional, one q'aire but different sample members provide data in different modes
 - Generally used to improve coverage, e.g., to the extent web *Rs* are younger than in-person (FTF) *Rs*, mixing these modes may improve age representation; may also improve RR
 - Sequential mixed mode: One sample, cross-sectional, sample members recruited in increasingly effective – and expensive – modes if previous contact attempts not successful
 - Generally used to increase response rates, e.g., mail, phone, in-person
- ❖ See de Leeuw, 2005, 2018 for more detail

Mixed-Mode Design: Data Collection (2)

- Switch modes within questionnaire: One sample; different modes for different parts of questionnaire for same *Rs*
 - e.g., FTF-ACASI
- Longitudinal mixed mode: One sample, multiple time points; same persons measured with different modes at different times
 - e.g., interview modes may help engage sample members in study; once *Rs* are committed to the study, a self-administered mode might be adequate
- Parallel (separate) mixed mode: Different samples, different modes
 - Typical in international surveys where some modes are more practical and common in some countries than others

❖ See de Leeuw, 2005 for more detail

Concurrent Mixed-Mode

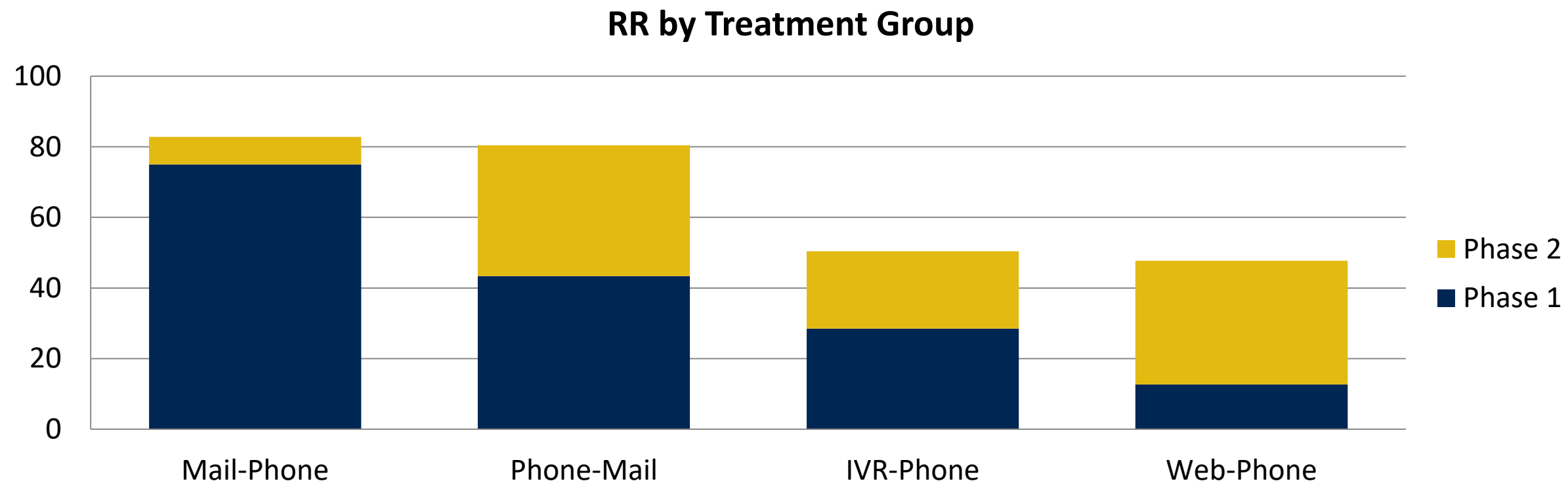
- Different modes for different subgroups of sample
 - e.g., Web for those with Internet access and device, mail for sample members without Internet access or device
 - Benefit: Increase coverage and response rate
 - Downside: Mode effects on measurement confounded with subgroups

Sequential Mixed-Mode Design (Nonresponse Follow-up)

- Goal: Reduce nonresponse
 - Use one mode as reminder for another mode
 - e.g., contact mail *nonrespondents* by phone to remind them to fill out paper questionnaire
- Goal: Reduce cost of nonresponse follow-up
 - Use increasingly more expensive modes for increasingly fewer, increasingly reluctant respondents
 - e.g., initially mail, then telephone, then face-to-face
 - Risk of mode effects on measurement: later modes used for increasingly reluctant sample members who might be less conscientious if participate

Mixed-Mode Design: Nonresponse Follow-up

- Dillman et al. (2009) followed up via Phone or Mail with nonrespondents to Mail, Phone, IVR, and Web invitations
 - Multiple attempts to collect data in initial mode
- Follow-up clearly increased response rates



Mixed-Mode Follow-up

- Main findings from Dillman et al. (2009)
 - Offering modes sequentially can improve RR
 - But does little to resolve demographic bias due to nonresponse: in general follow-up does not change demographic composition of original sample
 - Mode effect: *Rs* in voice modes (live phone interview, IVR) gave more positive responses on bipolar scales than in visual modes (mail, Web)

Switching Modes During Data Collection Session

- One sample, but different modes for different parts of questionnaire for same *Rs*
 - Use different modes to collect different types of data (e.g., sensitive versus not sensitive) from the same *Rs*
 - Improve data quality by increasing privacy of measurement
- Example: FTF + ACASI

Longitudinal Mixed Mode

- One sample, same persons/households in panel survey, measured with different modes at different times
 - Use one mode in first wave, another mode for later waves
 - Reduces costs, if start with more expensive mode
 - Confounds actual change over time with mode effects on measurement
 - e.g., PSID and CPS – use face-to-face for initial wave, mostly telephone thereafter
- Alternative: Collect email address in early wave(s) and invite to subsequent online waves via email

Parallel (Separate Samples) Mixed Mode

- Mainly used for comparative studies to accommodate different regional survey traditions or practical constraints (coverage, literacy, etc.)
 - Sometimes even different times and questionnaires
 - Mode effects may cause differences by country, state, or region, although many differences beside mode, including data collection organization
- Examples:
 - ISSP, ESS, BRFSS – different modes in different countries or states

Mode Choice

- Designers sometimes offer *Rs* a choice of data collection mode
 - Particularly well-suited to Concurrent Mixed Mode designs (usually no choice -- different frames used for different modes)
- Evidence suggests mode choice likely to reduce RR (relative to no choice) but improve data quality for those who do participate

Mode Choice: *Response Rate*

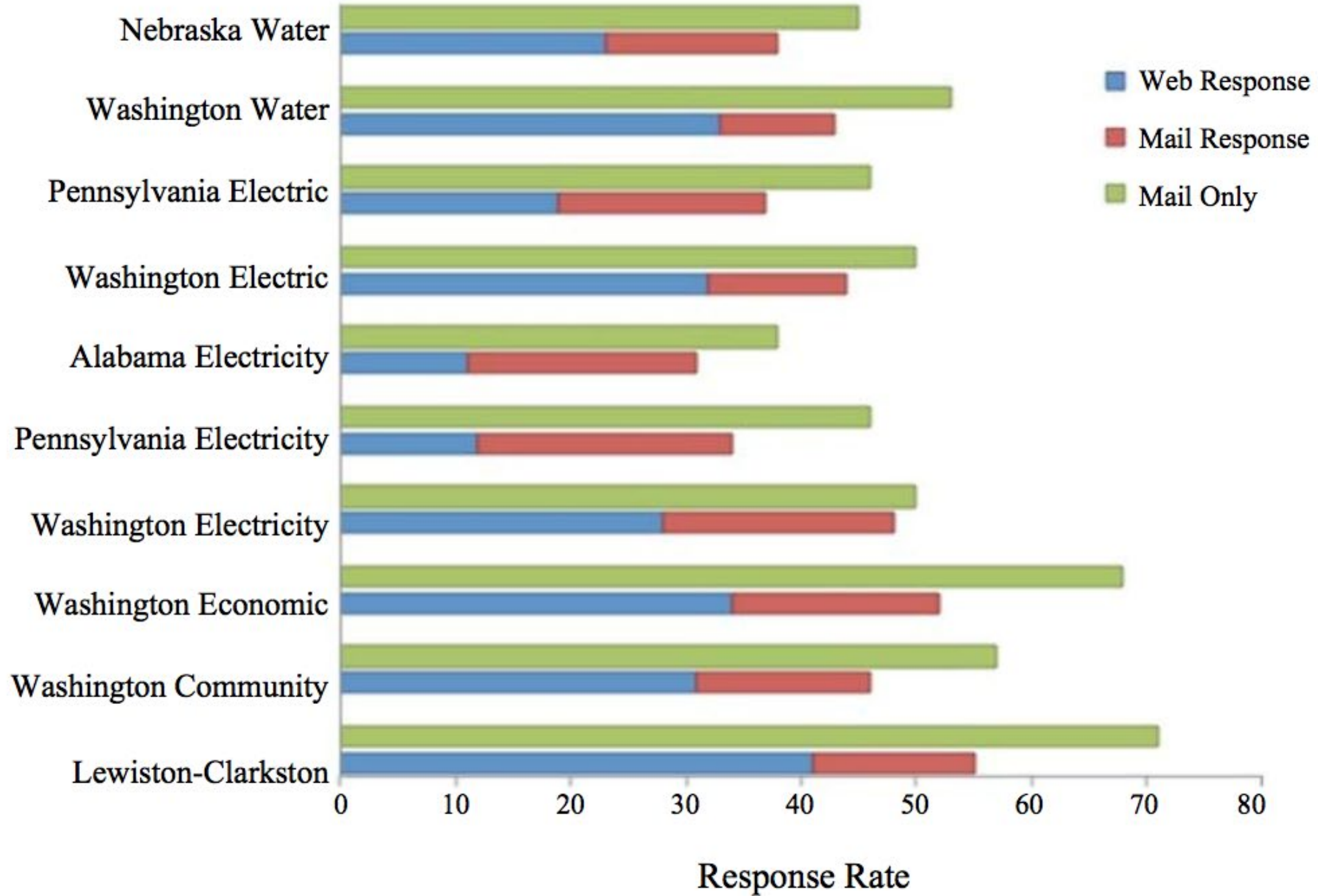
- Mode choice seems to reduce participation
 - Medway and Fulton (2012) meta-analysis
 - Reviewed 19 studies that mailed invitations to sample members and included paper questionnaire; random ½ Rs also offered web option
 - Choosing the web option requires typing the printed URL into device
 - Across the 19 studies, response rate was significantly lower overall in *choice* than *mail only* condition
 - RR lower in 17 out of 19 studies; significantly lower in 8

Mode Choice: *Response Rate* (2)

- Medway & Fulton propose that the negative impact of mode choice on response rate is due to:
 - 1.Paradox of Choice: simply being able to choose, makes both options less attractive than either alone
 - 2.Switching Costs: break in response process increases chances of losing URL or forgetting that intended to switch to web
 - 3.Usability/Connectivity problems (but the 3 studies that looked into this found little support for this with web completion > 80%)

Mode Choice: *Response Rate (3)*

- Push-to-web (Millar & Dillman, 2011): goal to maximize Web RR
 - Assumes there are benefits to Web over mail data collection
- In two experiments, postal mail invitation offering concurrent mode choice (Mail or Web) was expected to – but did not – improve RR beyond RR for Mail only
- But postal mail invitation offering concurrent choice (Mail or Web) followed up with email invitation that included link to web survey, led to RR comparable to Mail only
- Approach requires sampling frame with both primary email and street address for all sample members, so applicable to relatively narrow set of populations
- Dillman (2017) argues that initially contacting sample members via mail increases legitimacy of survey and allows inclusion of cash incentive; following with email invitation will produce RRs comparable to mail alone but with benefits of web
- In 10 tests, never quite produced equivalent mail-web and mail RRs but certainly shifted some respondents to web



Source: Dillman, 2019

Mode Choice: *Data Quality*

- Higher quality data when *Rs* choose mode (irrespective of mode)
- Conrad et al. (2017)
 - invited sample members in 1 of 4 modes :
Human Voice, Automated Voice, Human Text, Automated Text
 - Mode Choice Group: $\frac{1}{2}$ *Rs* were required to choose interview mode – could be same as contact mode or one of the other three
 - Assigned Mode Group: Other $\frac{1}{2}$ *Rs* assigned to participate in 1 of the 4 modes

Mode Choice Offer

“To get started, we need you to choose how you want to be interviewed -- whatever works best for you. There are four choices and any choice is fine with us. Do you want to ‘talk with a person’, ‘talk with an automated interviewer’, ‘text with a person’, or ‘text with an automated interviewer’?”

Mode Choice by Respondent: *Data Quality* (2)

- Mode Choice, compared to Assigned Mode, led to:
 1. Less rounding
 2. Less straightlining
 3. Fewer breakoffs
 4. Greater satisfaction with interview
- *Rs* could have chosen mode that requires least effort
- But *Rs*' choices do not seem driven by effort reduction: most common choice is Automated Text, enabling them to respond when have time to answer thoughtfully

Mode Choice: *Response Rate* (redux)

- Mode Choice led to slightly, but significantly, lower RR (52.1%) than Assigned Mode (55.9%), as in Medway & Fulton, 2012
- Not due to Paradox of Choice (Conrad et al. 2017):
 - RR no lower for *Rs* who chose to stay (participate) in invitation mode than for those assigned to a mode, i.e., choosing to stay in contact mode is a choice
 - but significantly lower for Mode Choice *Rs* who switched modes
- Suggests effect of choice on RR is largely due to Switching Costs

Mode Choice vs. Mode Preference

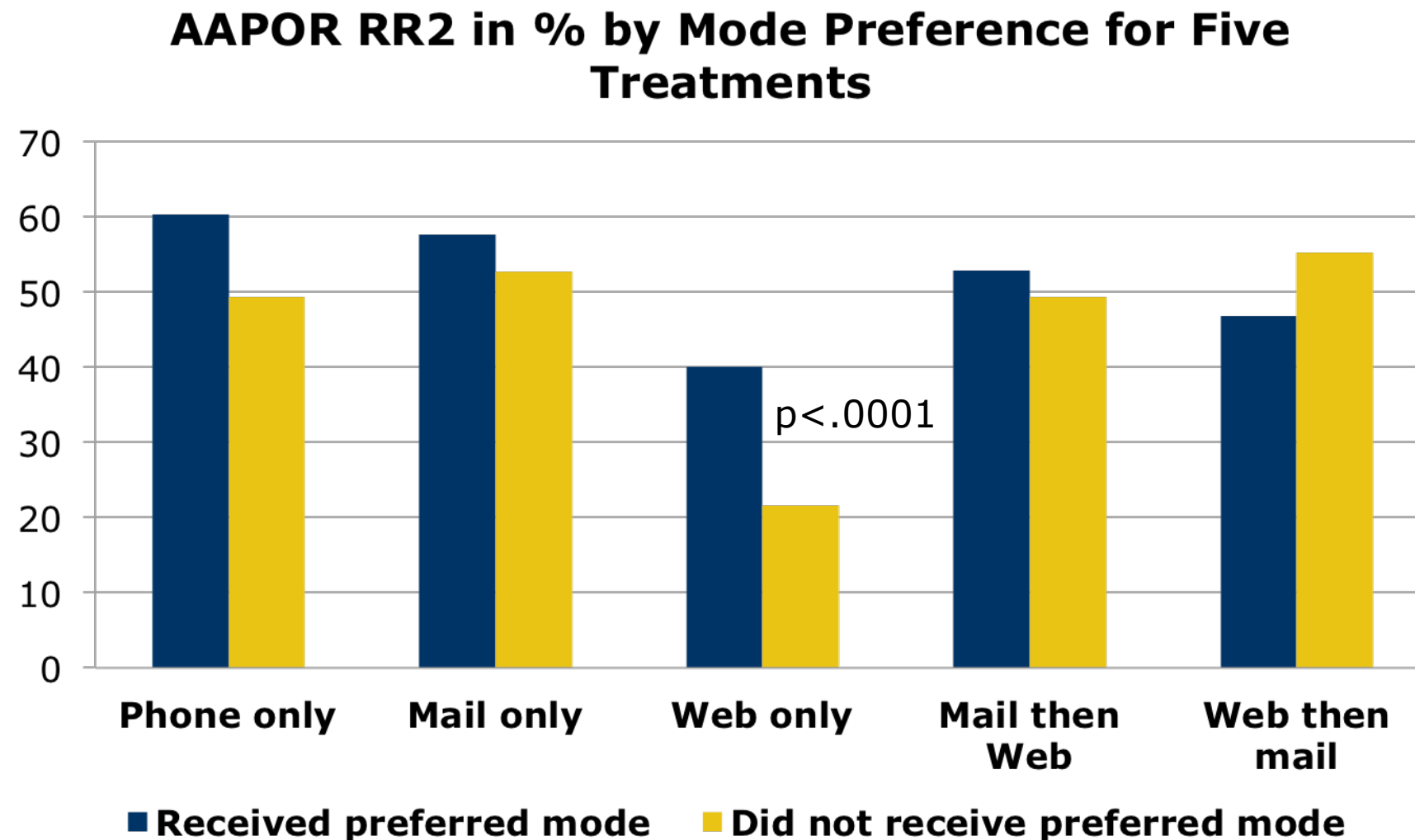
- Choosing a mode seems likely to motivate at least some *Rs* to participate and to do so conscientiously
 - But does seem to lower RR
- Inviting *Rs* to participate in a survey conducted in a mode they have previously indicated they prefer to other modes, may help improve data quality w/o reducing participation
- But may not be practical in most cross-sectional surveys because *Rs'* preferences are not known

Mode Preference: *Response Rate*

- Response rates higher if interviewed in preferred mode
 - Preferred mode $\neq$ chosen mode: *R*s may happen to participate in preferred mode but usually due to random assignment
 - Olson et al. (2012) asked *R*s their preferred mode in telephone survey:
If you received a request to do another survey like this one, would you prefer to participate in an in-person interview at your home, an interview on your home phone, an interview on your cell phone, a paper survey sent by mail, or a survey on the Internet?
 - One year later, same *R*s invited to participate in another survey; randomly assigned to mail or web modes (or a sequence of mail-web or web-mail):
 - Some *R*s answered in preferred mode, and some did not

Mode Preference for Data Collection Method

Response Rates in Olson et al., (2012): *Rs* invited to second survey and assigned to one of four modes – possibly their preferred mode measured a year earlier in telephone survey



Mode Preference: *Response Quality*

- Smyth, Olson & Kasabian(2014) examined *Rs'* effort when answering in preferred vs. non-preferred mode
 - Mode preferences collected after telephone interview
 - embedded in data collection reported in Olson et al. (2012)
 - *Rs* randomly assigned to one of two questionnaires in which item formats differed:
 - “Check-all-that-apply” vs. Forced Choice
 - Big vs. small text box for open response

Check-all vs. Forced Choice (Smith, et al., 2014)

29. Thinking about your personal finances, which of the following have you done in the past 12 months? Check all that apply.

- ☐ Delayed/canceled purchasing a home
- ☐ Delayed/canceled plans to buy a car
- ☐ Delayed/canceled plans to make a major household purchase
- ☐ Cut back on vacation spending
- ☐ Cut back on eating out
- ☐ Cut back on home internet access
- ☐ Cut back on home landline telephone services

29. Thinking about your personal finances, please indicate whether or not you have done each of the following in the past 12 months.

	<u>Yes</u>	<u>No</u>
Delayed/canceled purchasing a home	☐	☐
Delayed/canceled plans to buy a car	☐	☐
Delayed/canceled plans to make a major household purchase	☐	☐
Cut back on vacation spending	☐	☐
Cut back on eating out	☐	☐
Cut back on home internet access	☐	☐
Cut back on home landline telephone services	☐	☐

- *Rs* in non-preferred mode invest less effort in task
 - endorsed fewer options in “Check-all-that-apply” than Forced Choice and skipped open format *Qs* more with larger than smaller answer box
- If *Rs* responded in preferred mode, no effect of format

Summary/Discussion: Mixed Mode Designs

- Mixing modes to promote response rate and reduce coverage error is commonly used
 - Mode of Prenotification and Invitation vs. Data Collection
 - Different designs for mixing data collection modes, e.g., e.g., concurrent vs. longitudinal mixed modes
- Mode Choice is likely expected by sample members as number of communication modes proliferates
 - Although seems to reduce RRs, mode choice seems to increase data quality and satisfaction for those who complete interview
 - Reduction in RRs seems to be due to switching costs (changing modes)

Switching Modes within Interview, from Interviewer- to Self-Administration

- Many demonstrations that allowing/requiring *Rs* to self-administer sensitive questions on the device used by *Iwer* for non-sensitive questions increases disclosure of sensitive info (i.e., socially undesirable answers) compared to *Iwer*
- Computer Assisted Self-Administered Interviewing (CASI) and Audio CASI (ACASI) routinely used to collect sensitive info in personal visits
- Tourangeau & Smith (1996) compared disclosure with and without interviewer
 - In general, men report 2-4 times as many heterosexual sex partners as women; presumed due to opposite directions of social desirability bias by gender
 - Self-administration reduces gender-based asymmetry for # reported sex partners; presumed due to reduced social desirability
 - Self-administration increases some reports of illicit drug use
 - ACASI produced more disclosure (compared to in-person) than CASI

Does Audio Self-Administration Increase Disclosure?

- Immersive audio (e.g., headphones) seems to disconnect listeners from surroundings
 - When both audio and visual input, people seem to give more attention to the audio
 - e.g., mobile phones conversations while driving (in simulator) led to 2-fold increase in failure to detect traffic signs and, when signals detected, slower RTs (Strayer & Johnston, 2001)
- But Couper, Tourangeau & Marvin (2009) find no differences between ACASI and CASI (text-only) in reporting sensitive items in survey of fertility (NSFG)
 - May be due to non-use of audio (15% in NSFG pretest, 36% in NSFG Main Study) or incomplete playing of audio (*Rs* answer in 80% of play time for sound files)
 - No difference due to presence of others
 - No literacy advantage, at least those with lowest levels of education do not report differently in ACASI than CASI (text-only)

Impact of Third Parties During Self-Administered Questionnaires

- Does presence of household members, during self-administration reduce sense of privacy enough to reduce disclosure?
- Reports of drug and alcohol use by adolescents via CASI and paper SAQ affected by presence of parents (Acquilino et al. 2000)
 - Less reported marijuana and alcohol use when parents or siblings present
 - Effects of parents on marijuana use observed for paper SAQ but not CASI
 - For older *Rs*, cohabiting partners have no effect, but presence of children reduce disclosure

Does Computerization Enhance Benefits of Self Administration?

- Paper self-administered q'aires (SAQ) have been used for sensitive *Q's* in in-person interviews for decades
 - Common for *I* to leave stamped envelope with *R* to mail completed questionnaire to research organization (e.g., Aquilino, 1994)
- Tourangeau & Smith (1998) report greater advantage for CASI than SAQ (for some measures)

Computerization and Self-administration (2)

- Tourangeau & Smith (1998) report greater advantage for CASI than SAQ (for some measures)

Table 22.2. Mode Effects in the Women's Health Study

Experimental Group	Mean Reported Sex Partners			Percent Admitting Illicit Use
	Past Year	Past 5 Years	Lifetime	
Self-administered questions	1.72	3.88	6.54	40.9
Conventional (SAQ)	1.56	3.37	6.88	42.5
Computer assisted (CASI)	1.89	4.40	6.25	39.3
Interviewer-administered questions	1.44	2.82	5.43	40.7
Conventional (PAPI)	1.56	2.86	4.58	39.3
Computer assisted (CAPI)	1.36	2.79	6.27	42.2

Computerization and Self-administration (3)

- Meta-analysis (Gnambs & Kaspar, 2015) of 39 published articles with at least one comparison of disclosure in computerized (CASI/ACASI) vs. paper (SAQ) self-administration
 - 460 effect sizes compared
 - Predicted more disclosure in computerized self-administration than in paper SAQ because of greater anonymity, immersion, impartiality
- Odds ratio of disclosure in Computer to Paper: $OR = p_C / p_P$
- Overall, significantly more disclosure for computer than paper
- Effect increased as sensitivity increased:
 - measure of sensitivity available for 283 out of 343 effect sizes

Interactive Voice Response (IVR)

- Audio-recorded spoken *Qs* presented via phone
- Typically, *Rs* answer via touchtone data entry (TDE)
- but in some implementations, *Rs* speak their answers (Speech IVR)
- When used to collect sensitive information, called T-ACASI

IVR (cont'd)

1. Outbound* IVR

- a. *Lower* phones *R* and switches to automated system for sensitive *Qs*
 - In addition to “T-ACASI” also referred to as “Recruit and Switch”
- b. Automated system phones *R*; widely used in pre-election polling
 - “Robocalls”

2. Inbound* IVR

- a. *R* phones automated system

Disclosure in IVR

- Several studies demonstrate that *Rs* provide more stigmatized answers in IVR* than phone interviews
 - More sensitive sexual behaviors (Turner et al., 1996)
 - More illicit drug use (Gribble et al. 2000)
 - Lower levels of customer satisfaction (Tourangeau et al., 2002)

Breakoffs

- Interval during switch from *lwer* to IVR is vulnerable to breakoffs, especially as pause length increases
 - 18% (Gribble et al. 2000)
 - 68% when transfer = 10 sec (Dillman et al, 2002)
 - 7% during transfer (Tourangeau et al. 2002, exp. 4)
 - Engaging *Rs* before transfer by asking innocuous *Qs* reduced breakoffs (Tourangeau et al. 2002, exp. 4)
- Once IVR data collection begins, breakoffs higher than in comparable telephone interview
 - 39% after IVR interview started and especially high as iterated across more household members (Tourangeau et al. 2002, exp. 4)
- Breakoffs more likely when no Interviewer present (robocalls)

Human versus Digital Voice in IVR

- Possible recorded human voice will affect disclosure and breakoffs differently than Text-to-Speech (TTS)
- Couper, Singer, Tourangeau (2004) compared live telephone interviewer to recorded human voice to 2 TTS voices (one more and one less machine-like)
 - Recruit and switch IVR
 - Sensitive and non-sensitive *Qs*
- 23% breakoff rate (45% of those during transition) overall but no difference between IVR voice type
- No effect of IVR voice on disclosure; all three led to more disclosure than comparable telephone interview

IVR and Redirected Inbound Call Sampling (RICS)

- Millions of misdialed #s and calls to nonworking numbers made each month in US
- These calls sometimes “redirected” for marketing
- Also possible for surveys (RICS): misdialed calls redirected to IVR with callers’ consent
 - members of public initiate these calls, so survey organization not burdening caller additionally
- Levine, Krotki & Lavrakas (2017, 2019) found representation of demographic groups comparable to that in most dual frame RDD samples
 - More night owls than in typical RDD samples – likely a selection effect
 - By overrepresenting minority groups, may “correct” for underrepresentation in other frames
 - Proposed as “fit-for-purpose” solution, i.e., sample does not necessarily represent US population
- Some measurement error (significant, absolute differences from CPS benchmarks); some primacy effects
- Ethical concerns about redirecting calls
 - What if redirected call was to request roadside or medical assistance?

Speech IVR

- Usually, IVR requires answers to be entered via keypad, but some *Rs* may prefer to speak their answers which must then be recognized by system
- Generally, speech IVR system generates hypothesis about what user (*R*) said
- Johnston, et al., 2013, compared system's speech recognition to human coders' judgments before and after system confirmed interpretation with *R*
- Agreement:
 - Before system confirmed initial hypothesis: 88.9%
 - After system revised hypothesis: 95.6%
- Survey estimates (means and percentages) from system and coders very similar despite imperfect automated recognition

Speech IVR: Sample Dialogs

Johnston et al. 2013

Categorical



I: How often do you read the newspaper? 'Every day',
'a few times a week', 'once a week', 'less than once a week', or 'never'?

R: Every day

I: Got it.

Numerical



I: Thinking about the time since your eighteenth birthday (including the recent past that you've already told us about), how many male partners have you had sex with?

R: None

I: I think you said '9'. Is that right? Yes or No.

R: No

I: Thinking about the time since your eighteenth birthday (including the recent past that you've already told us about), how many male partners have you had sex with?

R: Zero

I: Thanks

First Hypothesis: "Nine"

Explicit Confirmation

Last Hypothesis: "Zero"

Last Annotation: "Zero"

Summary: Self-Administration within Interview

- Self-administration within interview consistently increases disclosure of sensitive information compared to *lwer*-administration
 - for at least some *Qs*
 - whether interview conducted in-person (FTF) or by telephone
- Self-administration:
 - Seems to increase sense of privacy and thus honesty of answering sensitive questions, relative to *lwer*-administration
 - When computerized seems to promote disclosure even more than on paper; audio-recorded questions may increase disclosure but results not definitive
 - CASI maybe less effective when third parties present
- IVR can increase disclosure of sensitive info but generally increases breakoffs
 - approaches such as RICS may complement conventional RDD sampling in IVR
 - Speech IVR is accurate enough to be judiciously used in production surveys

Mixed Mode Follow-up With Initial Incentive

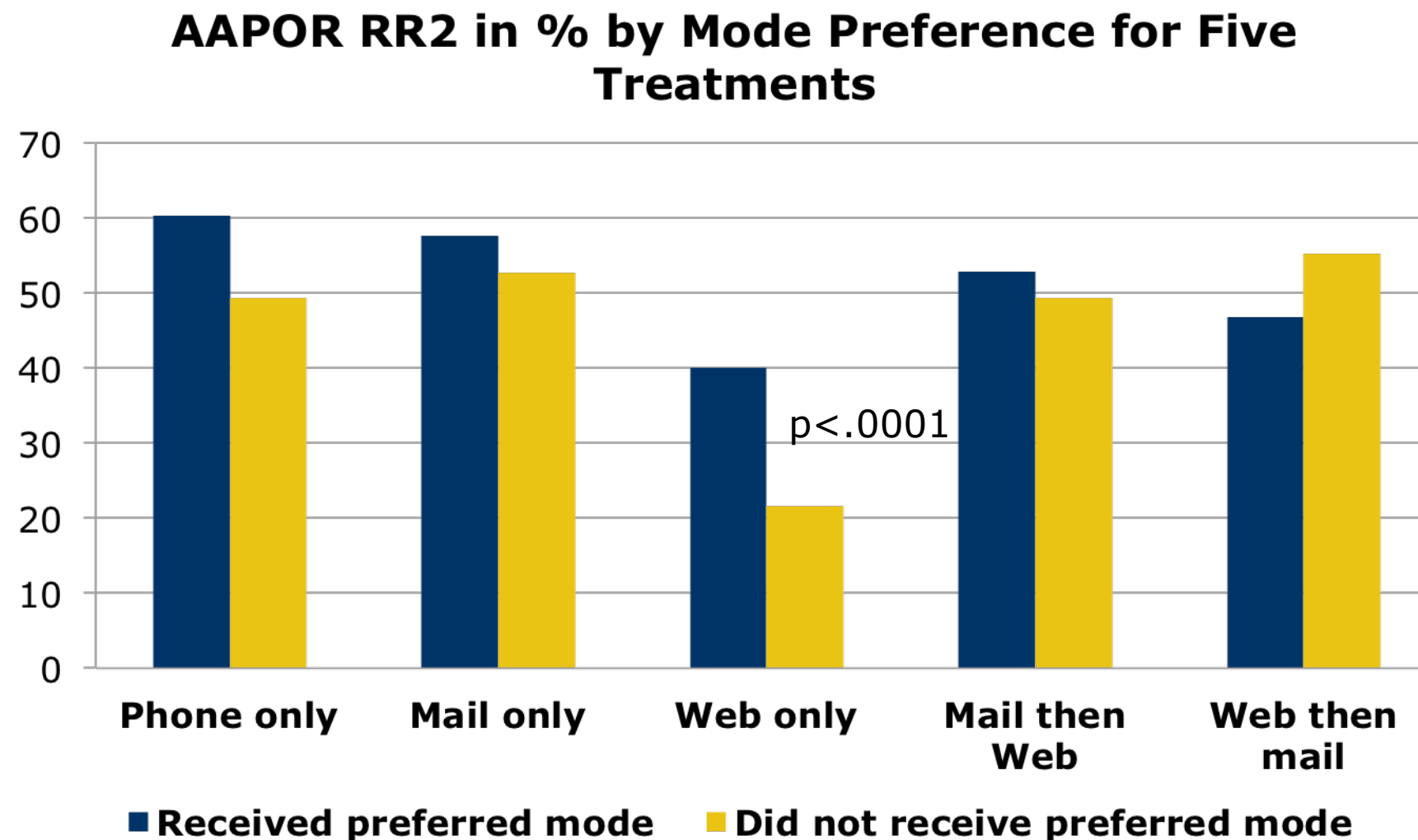
- Beebe et al. (2005) mailed q'aires to a sample of Medicaid recipients; ½ received \$2 incentive
 - RR with incentive = 54%, no incentive = 45%
- Telephone follow-up for minority/ethnic nonrespondents increased RR
 - RR with incentive = 69%, no incentive = 64%
- Effect of initial incentive reduced in follow-up
- Overall, increased RR with telephone follow-up offset cost of incentive

Mode Preference: *Response Rate*

- Response rates higher if interviewed in preferred mode
 - Preferred mode $\neq$ chosen mode: *R*s may happen to participate in preferred mode but usually due to random assignment;
 - Olson et al. (2012) asked *R*s preferred mode in telephone survey:
If you received a request to do another survey like this one, would you prefer to participate in an in-person interview at your home, an interview on your home phone, an interview on your cell phone, a paper survey sent by mail, or a survey on the Internet?
 - One year later, same *R*s invited to participate in another survey; randomly assigned to mail or web modes (or a sequence of mail-web or web-mail):
 - Some *R*s answered (initially) in preferred mode and some did not

Mode Preference for Data Collection Method

Response Rates in Olson et al., (2012) *Rs* invited to second survey and assigned to one of four modes – possibly their preferred mode measured a year earlier in telephone survey



Mode Preference: *Response Rate*

- Main findings from Olson et al. (2012):
 - Mode preference has some influence on survey participation, but is not uniform across modes
 - Weak evidence that offering preferred mode leads to slightly faster response
 - People who prefer Web or phone were as likely to participate in mail survey as those who preferred mail – mail seems to be robust to effects of mode preference
 - Offering Web option along with mail survey did not increase RR over mail only

Measuring Mode Preference

Mode preference by mode of data collection (in %)

Mode preference	Mode of Data Collection		
	Mail	Face-to-face	Telephone
Mail	76	11	24
Face-to-face	6	68	17
Telephone	1	6	44
No preference	17	15	15

Source: de Leeuw (1992), Table 5.13

Measuring Mode Preference

Mode preference by mode of data collection (in %)

Mode preference	Mode of Data Collection	
	Personal interviews in phone <i>HHs</i>	Telephone interviews in phone <i>HHs</i>
F2F interview	78.4	22.7
Telephone interview	1.7	39.4
Mailed questionnaire	16.9	28.1
Mixed opinion	1.5	1.8
Don't know	1.5	8.0
Total	100.0	100.0
Missing	14	69
Total N	1,437	1,696

ACASI vs. FTF I’wer-administration (CAPI)

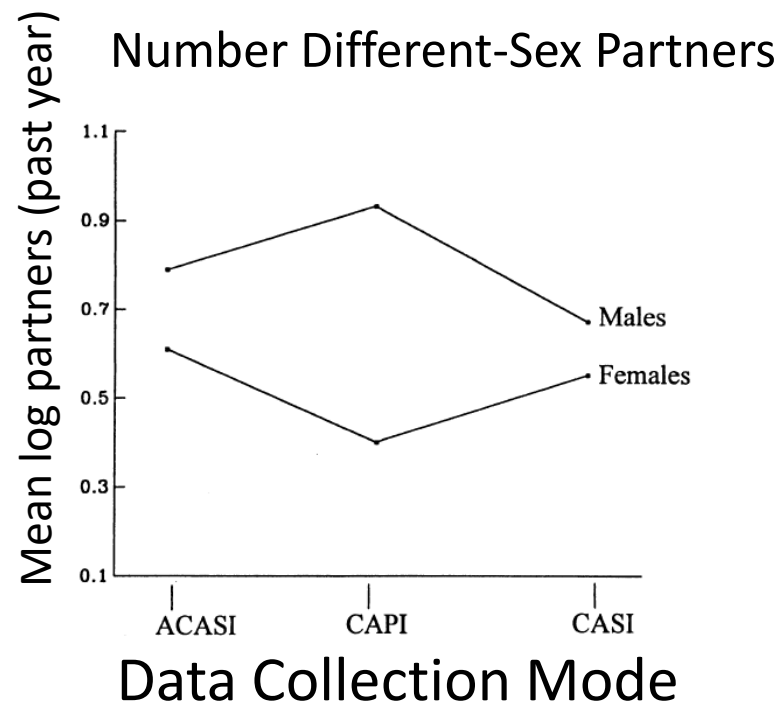
Dimension	Human interviewer	ACASI
Dialogue capability	Interviewer produces scripted utterances in real time in ways that will never be exactly the same for all respondents.	ACASI speech is prerecorded and each question is delivered in precisely the same way every time it is administered.
	Interviewers can, potentially, produce (scripted or unscripted) neutral probes or even clarification as needed.	ACASI systems do not generally produce probes or clarification (although in principle they could).
Embodiment	Interviewer’s body and face can move and express her cognitive and affective states while speaking as while listening to answers and entering them into the computer.	ACASI systems have no bodily or facial display (see Cassell & Miller, 2008, p. 165-166, Foucault, 2009, and von der Pütten et al., 2011, for discussion of potential displays by virtual interviewing systems of the future).
	By their very presence interviewers display both visual and auditory cues of their demographic characteristics—their age, ethnicity, gender, etc.	ACASI systems only have auditory cues of the recorded interviewer’s demographics.
Perceptual capability	Interviewer can observe the <i>respondent’s</i> physical characteristics and displays, and can make inferences from them (valid or not) about the respondent’s cognitive and affective states, the physical condition and circumstances, and demographic characteristics that might be relevant to interpreting responses.	ACASI system cannot observe or make inferences about the respondent (though in principle more advanced systems could; see Picard, 1998).
Evaluative capability	Interviewer has the ability to judge (positively or negatively) the sensitive behaviors that respondents report in the interview.	The ACASI system cannot judge.

Table 1. Lind, Schober, Conrad, Reichert (2013)

Self-administration Increases Reported Sensitive Info

(Tourangeau & Smith, 1996)

- Men report 2-4 times as many heterosexual partners as women; presumed due to differential social desirability for Males and Females
- But M-F Δ is reduced with CASI and ACASI



- More illicit drug use reported with self-administration (ACASI and CASI) than FTF (CAPI)
- Mode difference for marijuana use: significant for lifetime, marginally significant for past year
- Mode difference for cocaine use: significant for lifetime

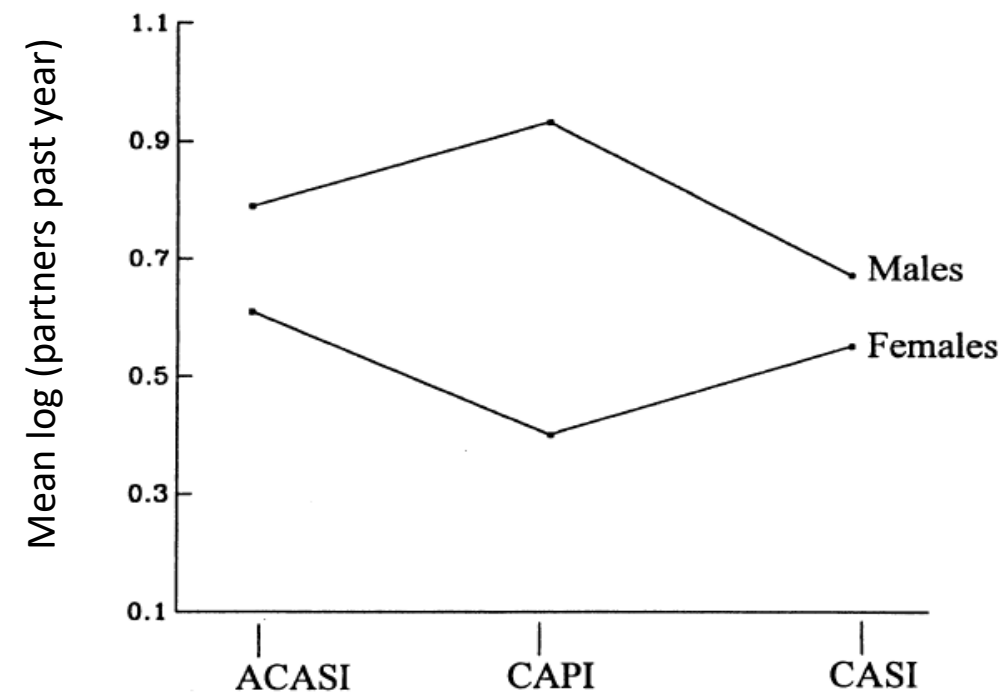
- May be extra benefit – more privacy -- from audio: More partners reported overall with ACASI

Asymmetry in ACASI: Bad faith or strategy use?

- Bogus pipeline study (T & S, 1998) suggests mode effects for sensitive Qs due to more and less honest reporting
 - Rs believe answers verified through polygraph-like machine
 - # sensitive behaviors on 15 of 19 Qs > than for control
- But, Brown and Sinclair (1999) suggest gender differences in sex partner Q due to gender-specific response strategies not social desirability
 - Males use “rough approximations” (general impressions) which lead to overestimation
 - episodes are hard to distinguish
 - Females enumerate partners which leads to underestimation
 - episodes are distinctive and memorable

Bad Faith or Strategy Use? (2)

- Possible both “motivated misreporting” and gender-specific strategies are involved
- Even under ACASI where social desirability should be low, some asymmetry remains



Paradox of computerization

- Depersonalizes and personalizes responding
 - CASI advantage presumably because computer alone lacks the social characteristics of a human intermediary
 - CASI and ACASI lack facial representation (Lind et al., 2013)
 - but “Computers as Social Actors” researchers (e.g. Nass & Moon, 2000) have shown that people ascribe human attributes to computers
 - Polite in evaluating the computer’s performance
 - Ascribe gender stereotypes to computers based on voice
- Which is it?

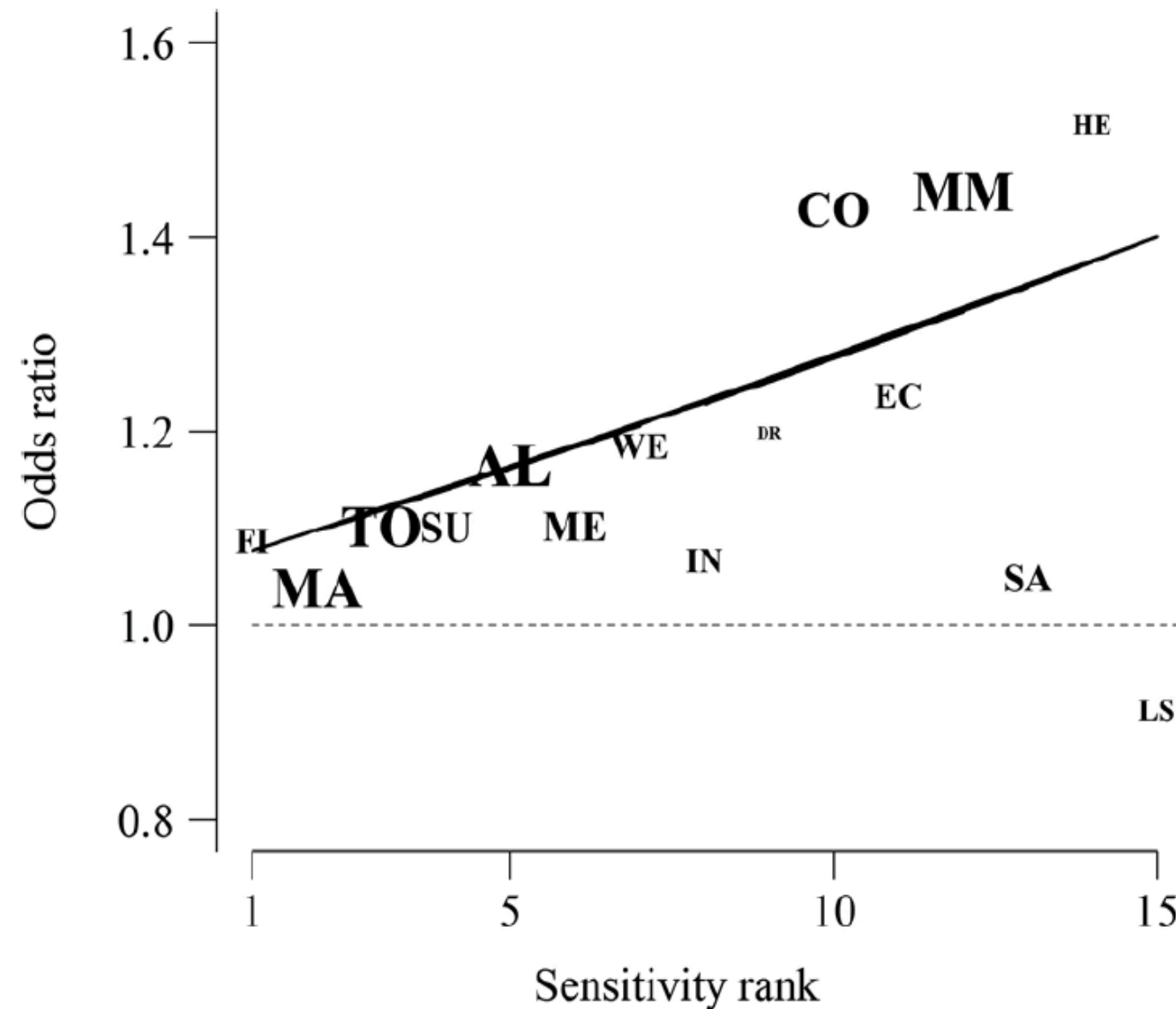
Social Presence in Self administration?

- Computers As Social Actors perspective implies interactions with computers in CASI should not feel particularly private
 - May be a particular concern on the web where privacy is already suspect
- “Humanizing” a web survey interface produced no evidence of socially desirable reporting (Tourangeau, Couper & Steiger, 2003)
 - humanized the interface by including picture of one of the male or female authors vs. logo
 - Varied interactivity by filling (or not) *R*’s name into questions
 - If CASA, then should be less disclosure in the humanized/interactive interface than the others
 - Only effect is gender of pictured author on gender attitudes, but no effect on disclosure of sensitive information

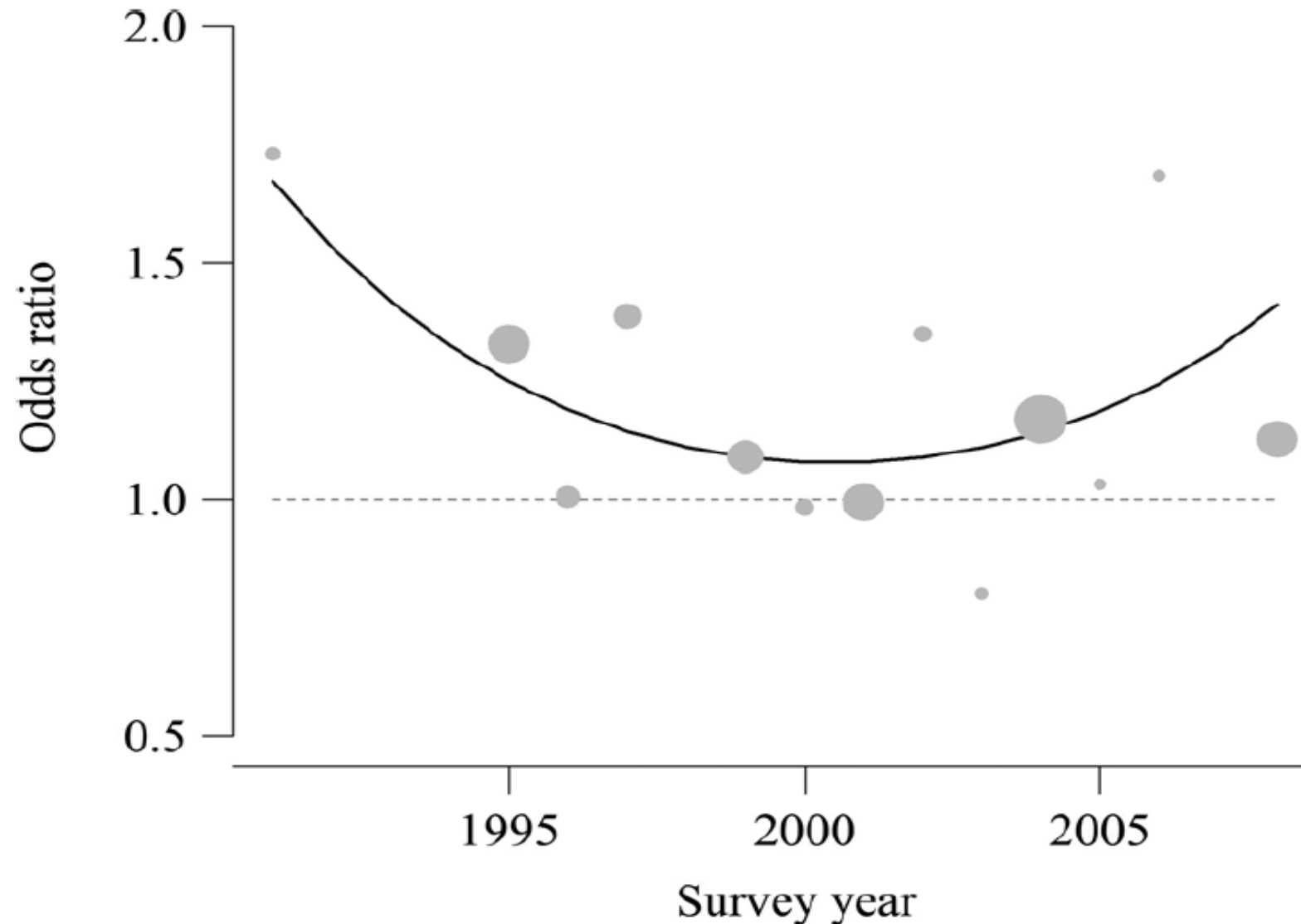
Social Presence in Self administration? (2)

- Possible resolution:
 - people do think of computers as agents and sometimes treat them like people
 - But they know the difference and recognize computers are different from people
 - Under certain circumstances, e.g. reporting sensitive behaviors, inanimacy and privacy of the medium outweighs its agent/human-like character
 - Stronger cues of animacy, e.g., facial movement and speech, may lead to social response as if /-administered (e.g., Lind et al., 2013).

Disclosure Advantage for Computer vs. Paper Self-Administration as function question sensitivity



Disclosure Advantage for Computer vs. Paper Self-Administration Over Time



Mode Preference: *Response Rate*

- Main findings from Olson et al. (2012):
 - Mode preference has some influence on survey participation, but is not uniform across modes
 - Weak evidence that offering preferred mode leads to slightly faster response
 - People who prefer Web or phone were as likely to participate in mail survey as those who preferred mail – mail seems to be robust to effects of mode preference
 - Offering Web option along with mail survey did not increase RR over mail only

Why do *Rs* disclose more in ACASI than FTF?

- Absence of human interviewer? Absence of human-like features such as a face?
- Can test by comparing disclosure in ACASI and FTF (extremes) to disclosure with virtual interviewers (*VIs*)
 - clearly automated but embodied
- Lind, Schober, Conrad & Reichert (2013) compared prevalence of undesirable answers in 4 modes:
 - (1) FTF, (2) High Animation (HA) Virtual Interviewer, (3) Low Animation (LA) Virtual Interviewer, (4) ACASI
 - T&S ('96) findings would suggest automated modes (ACASI, HA and LA) will promote disclosure relative to FTF interviews
 - But if virtual interviewers create *social presence* could inhibit disclosure, producing results more like FTF

Virtual Interviewer Interface

Lind, Schober, Conrad & Reichert (2013)



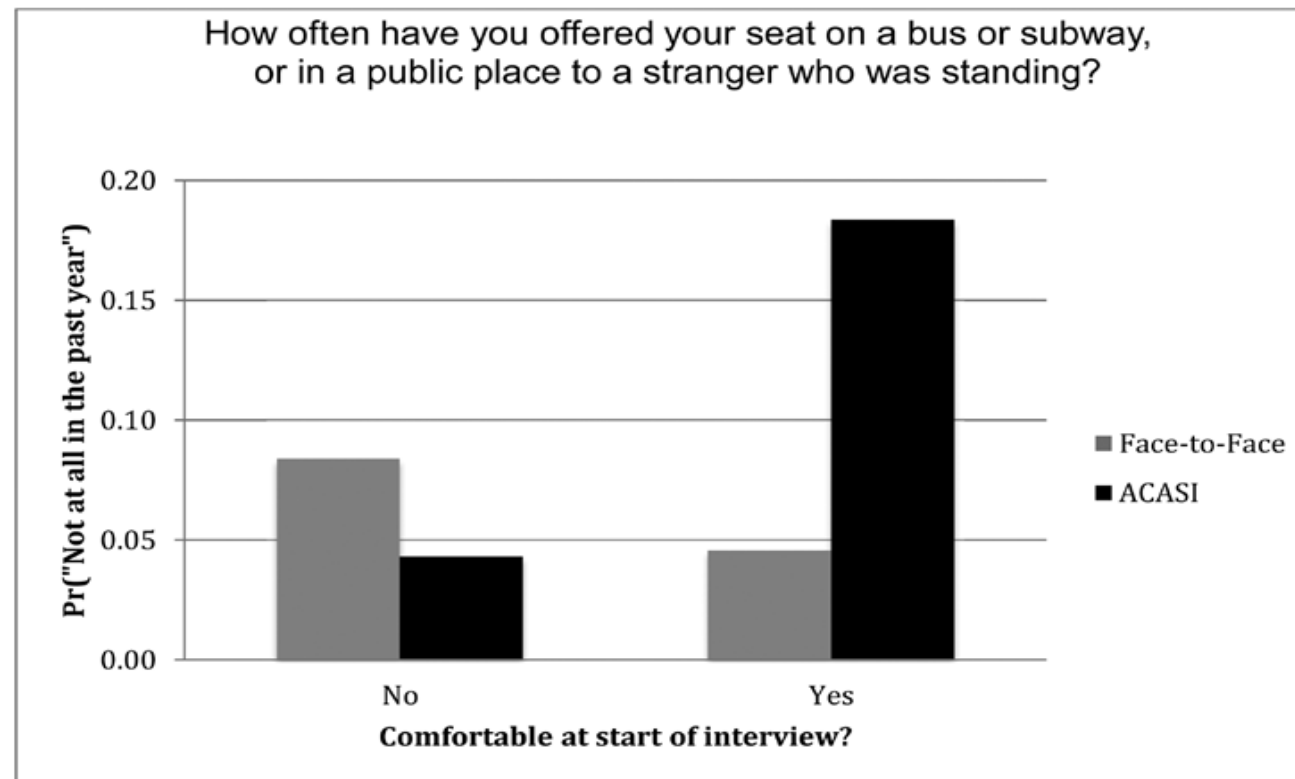
☐ Excellent
☐ Very good
☐ Good
☐ Fair
☐ Poor
☐ Don't know
☐ Not sure

[High Animation VI](#)

[Low Animation VI](#)

Lind et al. (2013) Main Findings

- Most common pattern: less disclosure in modes w/ than w/o a face
 - *Rs* disclose less in 8 of 10 questions in FTF, HA, & LA than ACASI
 - *R's* disclose less in FTF than the three automated modes (HA,LA, ACASI) for only 4 of 10 questions
- *Rs'* comfort affected impact of mode on disclosure



Speech IVR

- Although speech recognition is accurate, response rates lower in Speech IVR than Human-administered telephone interviews (Schober et al. 2012*)
 - 39% vs. 52%
- And percent “very satisfied” with interview is lower:
 - 45% vs. 66%
 - Although percent “very” and “somewhat satisfied” is the same: 92% vs. 95%

*same data set as Johnston et al. (2015)

Questions from Comments

- In **multi-mode mixed research**, what methods can be used to standardize the data?
- What are the best practices for ensuring respondent safety and privacy while maintaining data quality in mixed-mode surveys?
 - multiple online panels (both because respondents could answer with privacy and also because one panel alone didn't have a large enough trans sample). Additionally, there were many other methods employed to ensure respondent safety, including computer prompting such as "we understand this is a sensitive issue", or providing respondents with mental health resources, should they need them. Often, probability based panels provide reliable methods that are also sensitive to topics respondents may struggle with.
- Edith de Leeuw highlights the evolution and importance of mixed-mode surveys, which have been part of survey research **since the late 1980s**

Outline

- Mode effects in the context of mixed-mode surveys
 - Bias properties
 - How to control for nonrandom selection effects?
- Existing methods
- Mode comparison techniques

Survey Life Cycle for Mixed-Mode Surveys from a Quality Perspective (adapted from Groves et al., 2009)

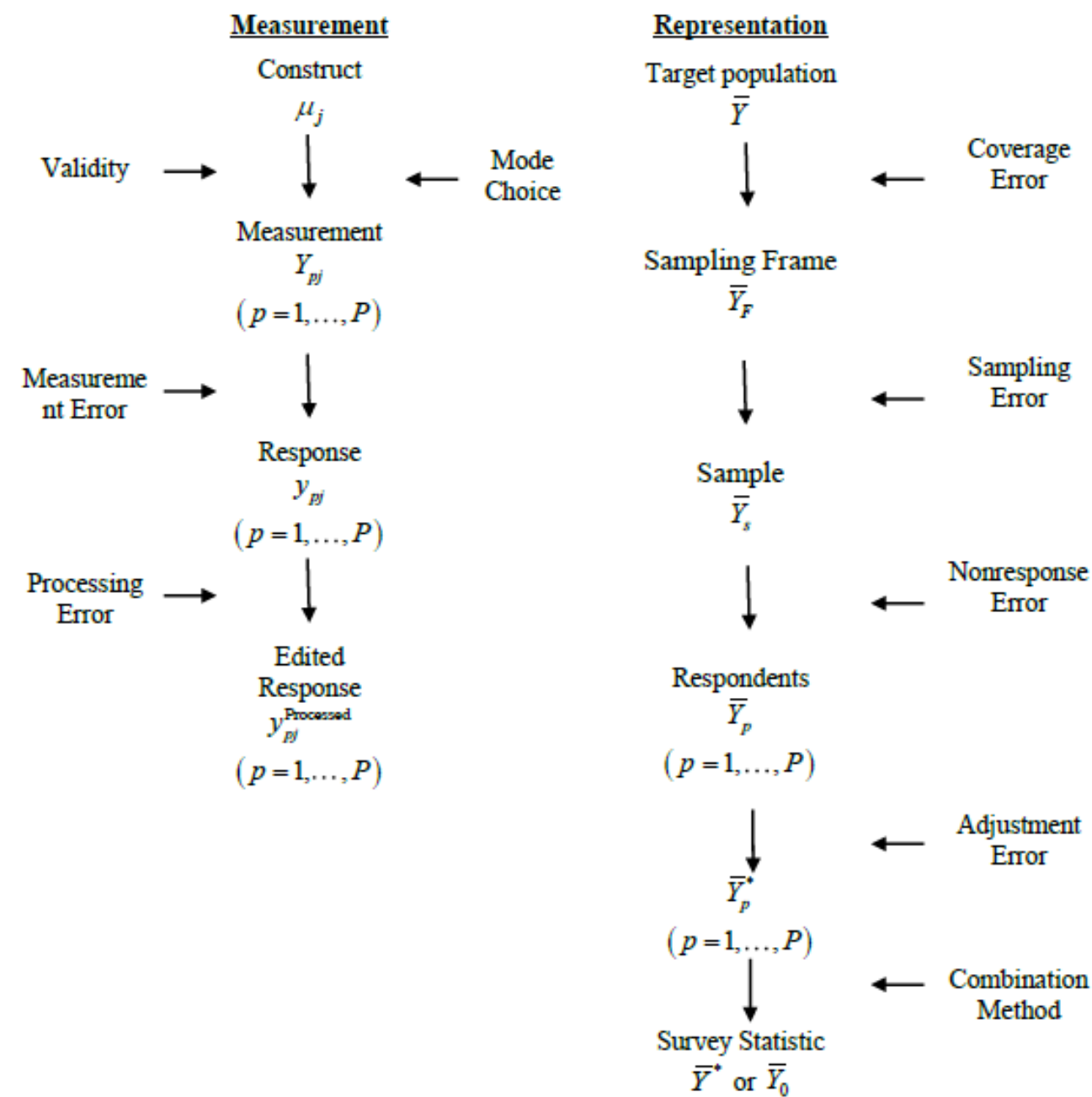


Figure 2.1 – Survey Life Cycle for Mixed-mode Surveys from a Quality Perspective

Survey Life Cycle for Mixed-Mode Surveys from a Quality Perspective (adapted from Groves et al., 2009)

The statistical notations in the survey life cycle in Figure 2.1 are:

μ_j : The true value of a construct for the j th population element

Y_{pj} : The measured value of a construct for the j th population element

collected by mode p

y_{pj} : The value of the response obtained for Y_{pj} collected by mode p

$y_{pj}^{\text{Processed}}$: The value of the response collected by mode p after editing
and other processing steps

$\bar{Y}$: Population mean of the Y_j 's

$\bar{Y}_F$: Population mean of interest for the part of the population covered
by the frame

$\bar{Y}_s$: Sample mean of interest

$\bar{Y}_p$: Mean of interest for respondents for a given mode p

$\bar{Y}_p^*$: Adjusted mean of interest for respondents for a given mode p

$\bar{Y}_0$: Mixed-mode survey mean that ignores mode effects

$\bar{Y}^*$: Mixed-mode survey mean

The estimator $\bar{Y}^*$ comes from combining the means from the different modes.

Statistical Error Model (Lee et al., 2016)

- Let's consider a particular observation under the influence of these errors under a stochastic mechanism. Let U be a population of size N , and the study goal be estimation of a mean μ for a population characteristic. For a unit j in U , the measurement process generates an observed value operationalized through y_j as:

$$y_{j,MEA} = m_j \mu_j$$

- Here measurement error, m_j , is a function of the measured variable itself (y_j) and a vector of parameters, say Ψ_{MEA} , in $m_j = f(y_j, \Psi_{MEA})$. Ψ_{MEA} may include observed auxiliary variables, and the function $f(y_j, \Psi_{MEA})$ can take any form (for example, linear or logistic regression model, generalized additive model).

Statistical Error Model

- For sample-based data collection, a sample S is drawn from a frame C , and sample units respond at less than a 100% rate. In the total error framework, frame coverage, sampling error, and nonresponse error play a role in observing the value of y_j in a series of conditional models:

$$y_{j,COV} = c_j y_{j,MEA}, \text{ where } j \in U,$$

$$y_{j,SAM} = s_j y_{j,COV}, \text{ where } j \in C, \text{ and}$$

$$y_{j,RES} = r_j y_{j,SAM}, \text{ where } j \in S.$$

c_j, s_j , and r_j are indicators of being covered in the frame, selected in the sample, and responding to the survey, respectively, functions of y_j and some parameters relevant to each of these sources of error.

Statistical Error Model

- For instance, the sample selection error is

$$s_j = f(y_j, \Psi_{SAM}) = \begin{cases} 1, & \text{if } j \in S \\ 0, & \text{otherwise.} \end{cases}$$

Coverage, sampling, and response indicators can be further expressed as respective probabilities or propensities $Pr(c_j = 1|y_j, \Psi_{COV}) = p_j^{COV}$, $Pr(s_j = 1|y_j, \Psi_{SAM}) = p_j^{SAM}$, and $Pr(r_j = 1|y_j, \Psi_{j,RES}) = p_j^{RES}$. In survey estimation the inverse of these probabilities may be used as adjustment weights for $y_{j,RES}$, the set of sample responses.

Statistical Error Model

- In sum, typical sample-based data collection observes can be expressed as

$$y_{j,RES} = p_j^{COV} p_j^{SAM} p_j^{RES} m_j \mu_j$$

- This model does not consider interactions among sources of error, such as between measurement and response, but interactions could be added. If p_j^{COV} , p_j^{SAM} and p_j^{RES} are correctly specified and estimated, then inferences based on $y_{j,RES}$ will be valid, regardless of how the sample is drawn.

Statistical Error Model

- In mixed-mode surveys, we also have mode choice:

$$y_{pj,RES} = R_{pj}y_{j,SAM}, \text{ where } j \in S \text{ and } p=1,\dots,P.$$

Statistical Error Model

- In single-frame mixed-mode surveys, measurement takes place via multiple modes.
- When the nonresponse indicator is replaced with mode specific indicators, the same logic can be used to derive mode-specific mean estimators.
- For specificity, consider a two-mode survey with T = telephone and I = personal interview.
- The mixed-mode design is conditioned on one phase with multiple modes.
- Here assume that overall response is complete and that nonresponse to one mode option (e.g. $R_{Tj} = 0$) implies response in an alternative mode (e.g. $R_{Ij} = 1$).

Statistical Error Model

$$y_j^* = R_{Tj}y_{Tj} + R_{Ij}y_{Ij}$$

$$j = 1, 2, 3, \dots, N$$

$$R_{Ij} = \begin{cases} 1 & \text{if population unit } j \text{ responds in in-person mode} \\ 0 & \text{if otherwise} \end{cases}$$

$$R_{Tj} = \begin{cases} 1 & \text{if population unit } j \text{ responds in telephone mode} \\ 0 & \text{if otherwise} \end{cases}$$

Statistical Error Model

- To reflect the possibility that the value reported when a person uses mode m may be incorrect, suppose that

$$y_{pj} = \mu_j + B_{pj} + \varepsilon_{pj} \quad (P = T \text{ or } I)$$

where ε_{pj} is a random error with mean 0. Consider the possible differences between y_{pj} and μ_j could be due to validity violations or/and measurement conditions that may vary by mode.

- B_{Tj} : reporting error for person j who responds by telephone,
- B_{Ij} : reporting error for person j who responds by in-person.

Statistical Error Model

- Letting E_M denote the expectation with respect to model , the average value reported by person j is $E_M(y_{pj}) = \mu_j + B_{pj}$, i.e., the report is biased compared to the desired value μ_j .
- Also, define U_p = set of persons that respond using mode p ($p = T$ or I).
- If we assume there is no undercoverage or overcoverage in the single-frame mixed-mode survey, then $U_S = U$. Given a particular split of the universe into U_T and U_I (i.e., conditioning on an outcome of the R_{pj} 's), the population estimator of the mean that does not correct for any differences in mode will be:

Statistical Error Model

- If we assume there is no undercoverage or overcoverage in the single-frame mixed-mode survey, then $U_S = U$. Given a particular split of the universe into U_T and U_I (i.e., conditioning on an outcome of the R_{pj} 's), the population estimator of the mean that does not correct for any differences in mode will be:

$$\bar{Y}^* = \sum_{p=(T,I)} \sum_{j \in U_p} \frac{y_j^* \lambda_k}{\eta_k}$$

Statistical Error Model

- Assume that the sample design is such that

$$\eta_k = E_S(\lambda_k)$$

- If $\overline{Y^*}$ was unbiased, its expectation (over all random quantities) would be $\bar{\mu} = \frac{\sum_U \mu_j}{N}$. However, if $B_{Tj}, B_{Ij} \neq 0$, i.e. in the presence of nonignorable mode effects, $\overline{Y^*}$ will be biased.

Statistical Error Model

- Considering model selection to be a random process, we can also model R_{pj} as a random variable with mean $E_R(R_{pj}) = g(x_j; \psi_p) \equiv g_{pj}$ where $g(\blacksquare)$ is a function like the logit or probit and x_j is a vector of covariates for person j . In the two-mode case with no nonresponse, assume that a single parameter vector ψ applies and that

$$E_R(R_{Tj}) = g(\mathbf{x}_j; \psi) = 1 - E_R(R_{Ij})$$

Statistical Error Model

- The order in which modes are made available to sample units can vary from one survey to another.
- If mode choice is modeled as random, this affects how one interprets the R_{pj} 's in .

Statistical Error Model

- For example, in a mixed-mode survey, response modes could be made available in a sequence or concurrently (Cobben, Schouten, & Bethlehem, 2006).
- In a sequential mixed-mode survey, the response modes that are made available to the sampled units vary depending on the phase. In an ACS-like design, in the first phase only the web response option is available. Sample units either respond by web or do not respond at all. In the second phase, when the mail mode is available, nonrespondents to the web mode can choose between web and mail modes to respond or do not respond at all.
- In contrast, in a concurrent Address Based Sample (ABS) Web-Mail survey, sample units can choose to select web or mail to respond or do not respond.

Statistical Error Model

- The statistical error model described in this section is used for three purposes in the following sections: (1) to compare the existing mixed-mode survey mode effect evaluation methods, and (2) to understand the bias properties of $\bar{Y}$.

Existing Mode Effect Evaluation Methods

- Current methods for evaluating mode effects focus on unconfounding the nonrandom selection of modes and mode effects in mixed-mode surveys.
- Regression model methods control for the nonrandom selection analytically (Jäckle et al., 2010).
- Mixture distribution method, introduced by Vannieuwenhuyze et al. (2010, 2012), computes the selection and mode effects for distribution parameter estimates, such as mean of a variable of interest, by defining θ as having a mixture distribution given the mixed-mode survey design.

Existing Mode Effect Evaluation Methods

- Using a parallel single mode survey and assuming the same population and measurement properties, mathematically it is possible to show that parameters related to the selection and the mode effects can be quantified for a two-mode survey.
- To do these comparisons, the mixture mode distribution method relies on two key assumptions:
 - single-mode survey and the mixed-mode survey are unbiased estimates for the same survey population with respect to the non-observational survey error.
- In practice, this could be a strong assumption as the mixed-mode surveys are usually conducted to minimize the nonresponse bias under a certain budget constraint.
- On the other hand, the method conceptualizes the confounding nature of mode choice and mode effects.
- Additionally, the method enables the computation of required sample sizes to detect mode effects.

Existing Mode Effect Evaluation Methods

- In practice, this could be a strong assumption as the mixed-mode surveys are usually conducted to minimize the nonresponse bias under a certain budget constraint.
- On the other hand, the method conceptualizes the confounding nature of mode choice and mode effects.
- Additionally, the method enables the computation of required sample sizes to detect mode effects.

Existing Mode Effect Evaluation Methods

- Alternatively, propensity score matching methods (Lee & Valliant, 2007; Rosenbaum & Rubin, 1983, 1984) unconfound the mode choice and the mode effects based on the propensity score matching strata that have been formed using the available covariates.
- This method defines the mode effects as the mean differences between the matched groups.

Regression Model Methods

- Jäckle et al. (2010) evaluate methods that define the mode effect as the differences in the mean or predicted response distributions between modes after controlling for some selected social-demographic variables.
- Although the data are obtained from randomized experiments, due to nonresponse the differences in the social-demographic distributions of respondents are controlled analytically.
- Jäckle et al. (2010) apply partial proportional odds models to test the linearity assumption for ordinal variables in addition to proportional odds and linear regression models.

Regression Model Methods

- The methods derive underlying response distributions for alternative modes conditioned on some selected social-demographic variables.
- These models are not linked to the social and the cognitive theories, they illustrate how different modeling assumptions may yield different results in mode effect evaluations.
- Jäckle et al. (2010) discuss two regression model structures in particular:

Regression Models

Regression Model for Continuous Variables

$$\hat{\mu}_p = X\hat{\beta}^{OLS} + R_p\hat{\beta}_p^{OLS}, \text{ where:} \quad (2.11)$$

$\hat{\mu}_p$: The predicted mean value for a given mode p , where

$p = 1, 2, \dots, P$,

X : Selected social-demographic covariates for respondents,

$\hat{\beta}^{OLS}$: Linear regression model parameter estimates associated with the social-demographics using ordinary least squares (OLS),

R_p : $N \times P$ matrix that includes dummy indicators for modes

$p = 1, 2, \dots, P$,

$\hat{\beta}_p^{OLS}$: Mode effect estimate under linear regression model.

Regression Models

Generalized Ordered Logit/Partial Proportional Odds Models for Ordinal Variables

$$\Pr(y_p > c) = \frac{\exp(X \hat{\beta}_c^{Odds} + R_p \hat{\beta}_{pc}^{Odds})}{1 + [\exp(X \hat{\beta}_c^{Odds} + R_p \hat{\beta}_{pc}^{Odds})]}, \text{ where:} \quad (2.12)$$

(2.12)

$\Pr(y_p > c)$: Predicted proportion of $y_p > c$ in which $c = 1, 2, \dots, C$ index the response categories,

$\hat{\beta}_c^{Odds}$: Generalized ordered logit or partial proportional odds model parameter estimates for selected social-demographic covariates for respondents for a given category,

$\hat{\beta}_{pc}^{Odds}$: Mode effect estimate under generalized ordered logit or partial proportional odds model for a given category.

Regression Models

- Generalized ordered logit models allow a relaxation of the assumption of the parallel line regression or proportional odds assumption that is generally implicit in modeling ordinal variables (Williams, 2006).
- The parallel line regression or proportional odds model assumption imposes the constraint that regression parameters except for the ordered intercepts to be same for all the response categories.

Regression Models

- Jäckle et al. (2010) showed that in assessing the mode effects this assumption may yield different conclusions.
- This finding is particularly important since it reiterates the importance of tying the social and the cognitive theories to the statistical models.
- In the follow-up research, Lynn, Hope, Jäckle, Campanelli, and Nicolaas (2011) use these models to test specific mode effect hypotheses based on the social and the cognitive theories.
- This kind of work will help to improve both the mixed-mode design principles and set the modeling assumptions in the adjustment models.

Mixture Distribution Method

- The method described in Vannieuwenhuyze et al. (2010, 2012) uses a single-mode survey as the reference distribution and estimates selection and mode effects for a two mode survey with respect to this reference distribution.
- This method considers mode-specific distributions of Y , one as measured by in-person (I) survey mode Y_I , and another as measured by telephone (T) survey mode Y_T when two survey modes are considered. While the method is not specific to one parameter of a distribution, the mean is considered for this discussion.
- $\bar{\mu}_U = \frac{1}{N} \sum_{j \in U} \mu_j$ is the parameter of interest given the underlying Y distribution.

Mixture Distribution Method

Assuming that one of the mode-specific distributions Y_I or Y_T can be measured, the density function of the mode-specific distribution can be written as a combination of conditioned distributions:

$$f_T(Y) = \bar{g}f_T(Y | U_T) + (1 - \bar{g})f_T(Y | U_I) \text{ , where:}$$

$$\bar{g} = \frac{N_T}{N} \text{ and, as in Section 2.2.2,}$$

U : is the population set,

U_T : set of telephone respondents,

U_I : set of in-person respondents,

$$U = U_T \cup U_I. \tag{2.13}$$

Mixture Distribution Method

Table 2.1 – Selection and Mode Effects as defined by Vannieuwenhuyze et al. (2010, 2012)

$Y_T U$	$Y_T U_T$	$Y_I U_I$
(1)	(2)	(3)
$\mu_T U_T$	$\mu_T U_T$	
$\mu_T U_I$		$\mu_I U_I$

Mixture Distribution Method

- As illustrated in Table 2.1, given a mixed-mode survey only conditional means can be observed for a given mode.
- The columns represent the conditional distribution of the variable of interest, Y .
- Column (1) represents a single-mode survey data that presumably collects survey data that represents all the members of the population using a telephone survey mode.
- Columns (2) and (3) together represent data collected by a mixed-mode survey.

Mixture Distribution Method

- Presumably, this mixed-mode survey data also represent all the members of the population but in addition to telephone interview data, Column (2), some data are collected by in-person interviews, Column (3).
- Rows (2) and (3) illustrate conditional means given a two mode survey.
- Shaded cells in Rows (2) and (3) cannot be observed in a mixed-mode survey.
- On the other hand, a single mode survey can produce the corresponding conditional means using one mode for these unobserved cells.

Mixture Distribution Method

$$\Pr(R_{T_j} = 1) = g_j \text{ and } \Pr(R_{F_j} = 1) = 1 - g_j \quad (2.14)$$

Instead of defining a function for the selection mechanism $g(\bullet)$ explicitly,

Vannieuwenhuyze et al. (2010, 2012) use $\bar{g} = \frac{N_T}{N}$ as shown in (2.13).

Mixture Distribution Method

With these definitions, the T selection effect is the difference in the mean for units that are actually enumerated by T and the mean for units actually enumerated by I but assuming that both sets of units used mode T :

$$S_T(Y) = (\bar{Y}_T | p = T) - (\bar{Y}_I | p = T) \quad (2.15)$$

The model-expectation under the model specified by (2.8) and (2.9) of the T selection effect is:

$$\begin{aligned} E_M [S_T(\bar{Y})] &= E_M [(\bar{Y}_T | p = T) - (\bar{Y}_I | p = T)] \\ &= \frac{1}{N_T} \sum_{j \in U_T} (\mu_j + B_{Tj}) - \frac{1}{N_I} \sum_{j \in U_I} (\mu_j + B_{Tj}) \\ &= \underbrace{\frac{1}{N_T} \sum_{j \in U_T} \mu_j}_{\bar{\mu}_T} + \underbrace{\frac{1}{N_T} \sum_{j \in U_T} B_{Tj}}_{\bar{B}_T(T)} - \underbrace{\frac{1}{N_I} \sum_{j \in U_I} \mu_j}_{\bar{\mu}_I} - \underbrace{\frac{1}{N_I} \sum_{j \in U_I} B_{Tj}}_{\bar{B}_I(T)} \quad (2.16) \\ &= \bar{\mu}_T + \bar{B}_T(T) - \bar{\mu}_I - \bar{B}_I(T) \\ &= [\bar{\mu}_T - \bar{\mu}_I] + [\bar{B}_T(T) - \bar{B}_I(T)] \end{aligned}$$

Mixture Distribution Method

The model-expectation of the in-person selection effect can be defined as the difference in the U_I and U_T sets, assuming that both were enumerated with mode I :

$$\begin{aligned} E_M[S_I(\bar{Y})] &= E_M[(\bar{Y}_I | p = I) - (\bar{Y}_T | p = I)] \\ &= \frac{1}{N_I} \sum_{j \in U_I} (\mu_j + B_{Ij}) - \frac{1}{N_T} \sum_{j \in U_T} (\mu_j + B_{Tj}) \\ &= \bar{\mu}_I + \bar{B}_I(I) - \bar{\mu}_T - \bar{B}_T(I) \\ &= [\bar{\mu}_I - \bar{\mu}_T] + [\bar{B}_I(I) - \bar{B}_T(I)] \end{aligned} \tag{2.18}$$

Mixture Distribution Method

$$M_T(\bar{Y}) = (\bar{Y}_T | p = T) - (\bar{Y}_T | p = I) \quad (2.19)$$

The model-expectation of telephone measurement effect is:

$$\begin{aligned} E_M[M_T(\bar{Y})] &= E_M[(\bar{Y}_T | p = T) - (\bar{Y}_T | p = I)] \\ &= \frac{1}{N_T} \sum_{j \in U_T} (\mu_j + B_{Tj}) - \frac{1}{N_T} \sum_{j \in U_T} (\mu_j + B_{Ij}) \\ &= \frac{1}{N_T} \sum_{j \in U_T} (B_{Tj} - B_{Ij}) \\ &= \bar{B}_T(T) - \bar{B}_T(I) \end{aligned} \quad (2.20)$$

Mixture Distribution Method

The mode effect could also be defined using the set of persons who responded in-person, i.e. U_I . Using similar notation to (2.19), we have

$$M_I(\bar{Y}) = (\bar{Y}_I | p = I) - (\bar{Y}_I | p = T) \quad (2.21)$$

The model-expectation of in-person measurement effect is:

$$\begin{aligned} E_M[M_I(\bar{Y})] &= E_M[(\bar{Y}_I | p = I) - (\bar{Y}_I | p = T)] \\ &= \frac{1}{N_I} \sum_{j \in U_I} (\mu_j + B_{\bar{Y}}) - \frac{1}{N_I} \sum_{j \in U_I} (\mu_j + B_{\bar{T}}) \\ &= \frac{1}{N_I} \sum_{j \in U_I} (B_{\bar{Y}} - B_{\bar{T}}) \\ &= \bar{B}_I(I) - \bar{B}_I(T) \end{aligned} \quad (2.22)$$

Mixture Distribution Method

Next, consider two hypothetical means—one in which the population mean is based entirely on responses via I and one in which the entire population responds by T . To that end, define

$\bar{\bar{Y}}_T = N^{-1} \sum_U (y_j^* | p = T)$ is the mean assuming that all persons respond by T

$\bar{\bar{Y}}_I = N^{-1} \sum_U (y_j^* | p = I)$ is the mean assuming that all persons respond by I

Calculating the model expectations of $\bar{\bar{Y}}_T$ and $\bar{\bar{Y}}_I$ gives

$$\begin{aligned} E_M(\bar{\bar{Y}}_T) &= N^{-1} \left[\sum_{U_T} (\mu_j + B_{Tj}) + \sum_{U_I} (\mu_j + B_{Tj}) \right] \\ &= \bar{g} [\bar{\mu}_T + \bar{B}_T(T)] + (1 - \bar{g}) [\bar{\mu}_I + \bar{B}_I(T)] \end{aligned}$$

and

$$\begin{aligned} E_M(\bar{\bar{Y}}_I) &= N^{-1} \left[\sum_{U_T} (\mu_j + B_{Ij}) + \sum_{U_I} (\mu_j + B_{Ij}) \right] \\ &= \bar{g} [\bar{\mu}_T + \bar{B}_T(I)] + (1 - \bar{g}) [\bar{\mu}_I + \bar{B}_I(I)] \end{aligned}$$

Mixture Distribution Method

- That is, the overall hypothetical marginal measurement effect for means can be written as a weighted difference of the two measurement effects described in and .
- Unlike the model-expectation of selection effects in the previous equations , the model expectation of mode effects are defined conditional on the same group of respondents and include differences in average mode specific bias.
- As a result of the mixture distribution properties, mathematically all the parameters required to define selection and mode effects can be derived.

Mixture Distribution Method

- However, estimating counterfactual terms requires specialized data sets. Vannieuwenhuyze et al. (2012) give conditions under which a single-mode reference survey can be used for estimation in parallel to a mixed mode survey. The method assumes *completeness* and *representativity*, i.e. complete response and no change in the measurement mechanisms between single mode (reference survey) and mixed mode survey.

Propensity Score Matching

- Another method to assess mode effects in mixed-mode surveys is the propensity score matching method (Lee & Valliant, 2007; Rosenbaum & Rubin, 1983, 1984).
- In the context of mixed-mode survey data analysis, the propensity score matching method relies on detecting groups with similar mode choice probability scores, in alternative response modes and compare mode effects as defined by differences in means, for example $Y_T - Y_I$, for a given group which has a $\bar{g}$ score on the average (Camillo & D'Attoma, 2011; Lugtig et al., 2011).

Propensity Score Matching

- This method does not assume an underlying measurement error, the mode effects are implicitly defined as the difference in the average systematic reporting errors between modes for a given matching group.
- This seems to be a straightforward adjustment for mode choice mechanism, the issue of unbalanced data presents itself as a problem as in the other propensity score matching method applications.
- Unbalanced data occurs when there are no matched cases given a for a given set of covariates. When there is a different coverage by frame, this is inherited in survey data (Lugtig et al. 2011).

Propensity Score Matching

- For example, in Lugtig et al. (2011) study households were invited by mail to take a web survey and households with registered landline telephone numbers were contacted by telephone.
- Inherently, the telephone frame excludes households with only non-landline telephones.
- So by definition web data include responses from the non-landline telephone households and a portion of the web responses cannot be matched to telephone responses due to coverage differences.

Propensity Score Matching

- In addition to coverage differences, some responses may not be matchable because of differential nonresponse.
- Both Lugtig et al. (2011) and Camillo & D'Attoma (2011) studies exclude unmatched data to make the evaluations of the mode effects.
- In general, imbalance data yield to more restricted modeling assumptions in the propensity score matching (Iacus, King, & Giuseppe Porro, 2012)

Calibration Estimator

Buelens and Van den Brakel (2011) extended the classical GREG estimator for the mean of y . Assuming a linear association between the variable of interest and a subset of covariates:

$$y_j = \mu(X_j, \beta^{(\mu)}) + \varepsilon_j, \text{ where:} \quad (2.24)$$

$j = 1, 2, 3, \dots, N$ indexes individual population persons,

μ_j can depend on X_j , a vector of covariates for person j ,

$$\varepsilon_j \stackrel{iid}{\sim} (0, \sigma^2).$$

Following the usual calibration notation, the corresponding GREG estimator for the total of Y , T , is

$$\hat{T}_{yr} = \sum_{j \in S} w_j y_j, \text{ where:} \quad (2.25)$$

$$w_j = \frac{1}{\pi_j} \left(1 + \frac{X_j^T}{\text{var}_M(y_j)} \left(\sum_s \left(\frac{X_j X_j'}{\text{var}_M(y_j) \pi_j} \right)^{-1} (T_X - \hat{T}_{X\pi}) \right) \right), \text{ where:}$$

π_j : the probability that unit j is included in sample S ,

$$T_X = \sum_{j \in U} x_j,$$

Calibration Estimator

$$y_j = \mu(X_j, \beta^{(\mu)}) + \varepsilon_j, \text{ where:} \quad (2.24)$$

$j = 1, 2, 3, \dots, N$ indexes individual population persons,

μ_j can depend on X_j , a vector of covariates for person j ,

$$\varepsilon_j \stackrel{iid}{\sim} (0, \sigma^2).$$

Following the usual calibration notation, the corresponding GREG estimator for the total of Y , T , is

$$\hat{T}_{yr} = \sum_{j \in S} w_j y_j, \text{ where:} \quad (2.25)$$

$$w_j = \frac{1}{\pi_j} \left(1 + \frac{X_j^T}{\text{var}_M(y_j)} \left(\sum_S \left(\frac{X_j X_j'}{\text{var}_M(y_j) \pi_j} \right)^{-1} (T_X - \hat{T}_{X\pi}) \right) \right), \text{ where:}$$

π_j : the probability that unit j is included in sample S ,

$$T_X = \sum_{j \in U} x_j,$$

$\hat{T}_{X\pi}$: the π estimator of T_X .

Mode Effects

- Single-mode surveys:
 - Differences in overall results: Mode effects are part of trade-off analysis, no assumption about the ignorability of mode effects
- Mixed-mode surveys:
 - Assumption: Ignorable mode effects
 - Social desirability bias: Respondents are more likely to misreport their statuses on sensitive topics conditioned on their status in the presence of an interviewer (Tourangeau & Smith, 1996; Tourangeau & Yan, 2007)
 - In-person respondents may be more immune to social desirability tendencies (Holbrook, Green, & Krosnick, 2003)
 - E.g., Income is a sensitive topic in the U.S. (Moore, Stinson & Welniak, Jr, 2000), also an observable characteristic

Mode Effects in Mixed-Mode Surveys

- Mode choice: Nonrandomized Mode Assignment
 - E.g., Respondents with higher education are more likely to respond in telephone mode than in in-person compared to respondents who have less than a 12th Grade education (CPS, March 2012)
- Mode effects are confounded by mode choice in mixed-mode surveys

Measurement Error Model for a Mixed-Mode Survey

$$y_j = \mu(X_j, \beta^{(\mu)}) + R_{jT} B_{jT} + R_{jI} B_{jI} + \varepsilon_j$$

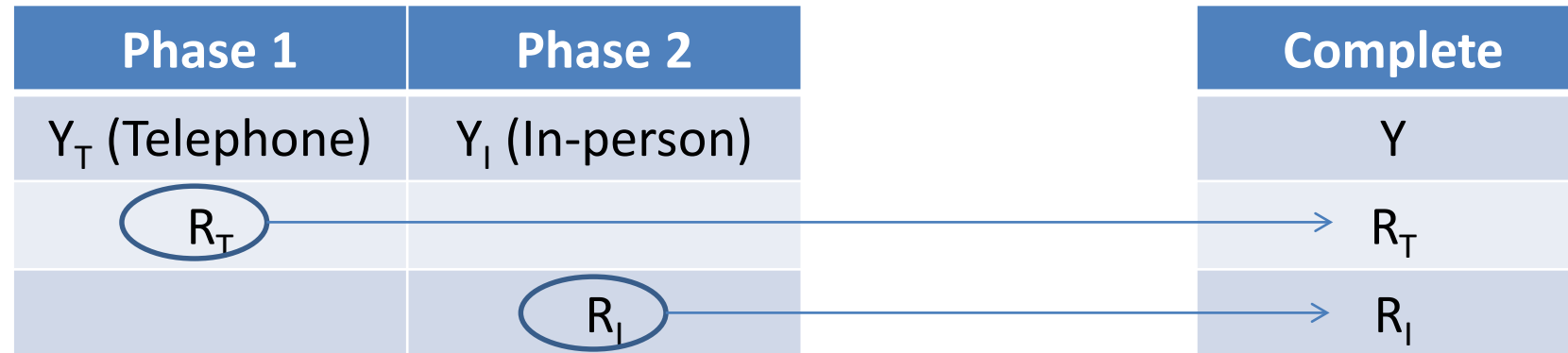
Mode choice

$$\left\{ \begin{array}{l} R_{jT}, R_{jI} : \text{indicator variables for response mode} \\ E_R(R_{jT}) = g(X_j; \psi) \equiv g_j \end{array} \right.$$

Mode effects

$$\left\{ \begin{array}{l} \mu_j = \mu(X_j, \beta^{(\mu)}) \\ B_{jT}, B_{jI} : \text{mode effects} \\ \text{Alternatively, } B_{jT} = Z_j^T B_T, \quad B_{jI} = Z_j^T B_I \end{array} \right.$$

Population Mean Ignoring Mode Effects



$$\bar{Y}_0 = \frac{1}{N} \left[\sum_{j \in U_T} y_j + \sum_{j \in U_I} y_j \right]$$

U_T = Telephone respondents

U_I = In-person respondents

$$E_M[\bar{Y}_0 - \bar{Y}] = \overline{(gB)}_U$$



a weighted average of
telephone and in-
person mode effects